



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143-02 Probability & Statistical Data Analysis

Assignment 3

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Assignment 3

Question 1

$$(a) \mu = \frac{\sum x}{n} = \frac{345+420+390+410+380+400+430+360+375+390+405+415+400+385+395}{15}$$
$$= \frac{5900}{15}$$
$$= 393.333$$

(b) Confidence interval -95%, ($\alpha=0.05$)

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n} = \frac{2335.789 + 711.129 + 11.109 + 277.789 + 177.769 + 44.449 + 1344.469 + 111.089 + 336.099 + 11.109 + 136.119 + 469.459 + 44.449 + 69.439 + 2.779}{15-1}$$
$$= \frac{7083.045}{14}$$
$$= 505.932$$

$$\sigma = \sqrt{505.932} = 22.493,,$$

$$393.333 \pm 2.145 \left(\frac{22.493}{\sqrt{15}} \right) = (380.873, 405.790),,$$

Question 2

$$(a) H_0: \mu = 5$$

$$H_1: \mu < 5$$

$$(b) Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$
$$= \frac{4.6 - 5}{1.2 / \sqrt{500}}$$
$$= -7.454$$

$$(c) n=500 \quad \sigma=1.2 \quad \bar{x}=4.6 \quad \alpha=0.01$$

$$Z_{critical} = -2.325$$

$$(d) \text{ Since } Z < Z_{critical} \quad (-7.454 < -2.325)$$

Reject H_0 .

(e) Reject H_0 . There is sufficient evidence to support the claim that the average waiting time for the customers in the bank's queue is less than 5 minutes ($\mu < 5$).

Question 3

$$\mu = 160 \quad n = 25 \quad \bar{x} = 183 \quad s = 12 \quad \alpha = 0.05$$

$$H_0 : \mu = 160$$

$$H_1 : \mu > 160$$

$$\text{Test statistic : } t = \frac{\cancel{180} - 160}{\frac{12}{\sqrt{25}}} = \frac{183 - 160}{12/\sqrt{25}}$$
$$t = 9.583$$

$$\text{Critical value : } t_{0.05, 24} = 1.711$$

$\therefore t \geq t_{0.05, 24}$ Thus, H_0 is rejected. There is sufficient evidence that the mean score for the students from this university is greater than 160.

Question 4

$$\mu = 50 \quad n = 38 \quad \bar{x} = 51 \quad \alpha = 0.01 \quad \sigma = 3.6$$

$$H_0 : \mu = 50$$

$$H_1 : \mu > 50$$

$$\text{Test statistic : } z = \frac{51 - 50}{\frac{3.6}{\sqrt{38}}}$$
$$z = 1.712$$

$$\text{Critical value : } z_{0.01} = 2.33$$

$\therefore z < z_{0.01}$. Thus, fail to reject H_0 . There is no sufficient evidence that the population mean is greater than 50 cm.

Question 5

$$H_0: \sigma = 7460$$

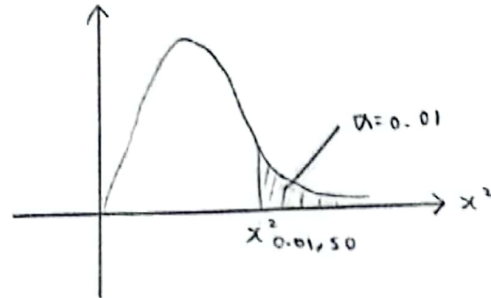
$$H_1: \sigma > 7460$$

$$\begin{aligned}\chi^2 &= \frac{(n-1)s^2}{\sigma^2} \\ &= \frac{(51-1)(7371)^2}{7460} \\ &= 55.68\end{aligned}$$

$$\alpha = 0.01$$

$$df = 51 - 1 = 50$$

$$\chi^2_{0.01, 50} = 76.154$$



Since $55.68 < 76.154$, fail to reject H_0 .

At a significance level of 0.01, there is not enough evidence to conclude that males have a greater standard deviation than females.

Question 6

$$\begin{aligned}\bar{x}_f &= \frac{4.3 + 4.3 + 4.4 + 4.0 + 3.4 + 4.7 + 2.9 + 4.0 + 4.3 + 3.4 + 3.4 + 3.3}{12} \\ &= 3.867\end{aligned}$$

$$\begin{aligned}\bar{x}_m &= \frac{4.3 + 3.7 + 4.2 + 3.9 + 3.1 + 4.0 + 3.8 + 3.4 + 4.5 + 3.8 + 4.3 + 4.4 + 4.1 + 4.2 + 4.0 + 4}{16} \\ &= 4.019\end{aligned}$$

female:

$ x - \bar{x} $	$(x - \bar{x})^2$
0.433	0.187
0.433	0.187
0.533	0.284
0.133	0.018
0.467	0.218
0.833	0.694
0.967	0.935
0.133	0.018
0.433	0.187
0.467	0.218
0.467	0.218
0.567	0.321
$\Sigma = 3.485$	

male:

$ x - \bar{x} $	$(x - \bar{x})^2$
0.481	0.231
0.319	0.101
0.181	0.033
0.119	0.014
0.919	0.845
0.019	3.61×10^{-4}
0.219	0.048
0.619	0.383
0.481	0.231
0.219	0.048
0.281	0.079
0.381	0.145
0.081	6.561×10^{-3}
0.181	0.033
0.019	3.61×10^{-4}
0.381	0.145

$$\Sigma (|x - \bar{x}|)^2 = 2.393$$

$$S_f = \sqrt{\frac{3.485}{12-1}}$$

$$= 0.563$$

$$S_m = \sqrt{\frac{2.343}{16-1}}$$

$$= 0.395$$

(b) $H_0 : \mu_1 = \mu_2$
 $H_1 : \mu_1 \neq \mu_2$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(12-1)(0.563)^2 + (16-1)(0.395)^2}{12 + 16 - 2}$$

$$= 0.224$$

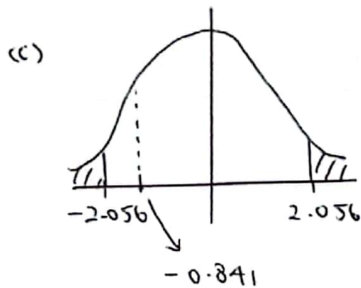
$$S_p = \sqrt{0.224} = 0.4734$$

$$t = \frac{\bar{x} - \bar{x}_2}{S_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{3.867 - 4.019}{0.4734 \sqrt{\left(\frac{1}{12} + \frac{1}{16}\right)}}$$

$$= -0.841$$

$$t_{0.025, 26} = 2.056$$



Fail to reject H_0 . There is not enough evidence to claim that two samples have different mean.

(d) $H_0 : S_1 = S_2$
 $H_1 : S_1 > S_2$

$$F = \frac{S_1^2}{S_2^2} = \frac{(0.395)^2}{(0.563)^2}$$

$$= 0.492$$

$$F_{0.01, 15, 11} = 4.25$$

$$0.492 < 4.25$$

$\therefore$ Fail to reject H_0 . At a significance level of 0.01, there is not enough evidence to claim that the variance of evaluation score of female professors is higher than male professors.

Question 7

Mother	Daughter	D	D ²
68.0	68.5	-0.5	0.25
60.0	60.0	0	0
61.0	63.5	-2.5	6.25
63.5	67.5	-4	16
69.0	68.0	1	1
64.0	65.5	-1.5	2.25
69.0	69.0	0	0
64.0	68.0	-4	16
63.5	64.5	-1	1
66.0	63.0	3	9
		$\sum D = -9.5$	$\sum D^2 = 51.75$

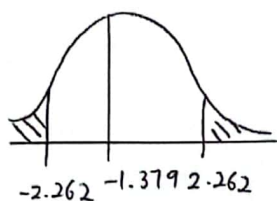
(b) $\bar{D} = \frac{-9.5}{10} = -0.95$

$$s_D = \sqrt{\frac{51.75 - \frac{(-9.5)^2}{10}}{9}} = 2.179$$

(c) $H_0 : \mu_D = 0$
 $H_1 : \mu_D \neq 0$

$$t = \frac{\bar{D} - \mu_0}{\frac{s_D}{\sqrt{n}}} = \frac{-0.95 - 0}{\frac{2.179}{\sqrt{10}}} = -1.379$$

$t_{0.025, 9} = 2.262$



Since $2.262 > -1.379 > -2.262$, fail to reject H_0 . At 0.05 significant level, there is not enough evidence to claim that there is no difference in height between mothers and their first daughters.

Question 8

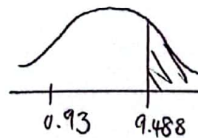
(a)

Cell, i	O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$
Email, Satisfied	90	$\frac{200 \times 200}{430} = 93.02$	$\frac{(90 - 93.02)^2}{93.02} = 0.098$
Email, Neutral	80	$\frac{200 \times 170}{430} = 79.07$	$\frac{(80 - 79.07)^2}{79.07} = 0.011$
Email, Dissatisfied	30	$\frac{200 \times 60}{430} = 27.91$	$\frac{(30 - 27.91)^2}{27.91} = 0.157$
SMS, Satisfied	70	$\frac{150 \times 200}{430} = 69.77$	$\frac{(70 - 69.77)^2}{69.77} = 7.582 \times 10^{-4}$
SMS, Neutral	60	$\frac{150 \times 170}{430} = 59.30$	$\frac{(60 - 59.30)^2}{59.30} = 8.263 \times 10^{-3}$
SMS, Dissatisfied	20	$\frac{150 \times 60}{430} = 20.93$	$\frac{(20 - 20.93)^2}{20.93} = 0.0413$
Phone call, Satisfied	40	$\frac{80 \times 200}{430} = 37.21$	$\frac{(40 - 37.21)^2}{37.21} = 0.209$
Phone call, Neutral	30	$\frac{80 \times 170}{430} = 31.63$	$\frac{(30 - 31.63)^2}{31.63} = 0.084$
Phone Call, Dissatisfied	10	$\frac{80 \times 60}{430} = 11.16$	$\frac{(10 - 11.16)^2}{11.16} = 0.121$
			$\chi^2 = 0.73$

- (b) H_0 : No association between customer satisfaction levels with banking services and their preferred method of communication for receiving updates from the bank.
 H_1 : Significant association between customer satisfaction levels with banking services and their preferred method of communication for receiving updates from the bank.

$$df = (3-1)(3-1) = 4$$

$$\chi^2_{4,0.05} = 9.488$$



Since $0.73 < 9.488$, fail to reject H_0 . There is not enough evidence to claim that significant association between customer satisfaction levels with banking services and their preferred method of communication for receiving updates from the bank.